

Changyuan Chen

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Education & Research Experience

Eastern Michigan University, USA	B.S. Computer Science	2024.09-2028.06
Tsinghua University (Prof. Yanyan Lan)	Research Intern, RL	2026.06-2026.09
Tencent Rhino-Bird Elite Program	Open-Source Agent Systems	2026.02-2026.07
❖ GPA: 3.93 (Top 0.4%, Rank 1/213) · English: IELTS 6.5, GRE 318		
❖ Major Courses: Artificial Intelligence and Computer Vision (94), Data Structures (95), Operating Systems (99), Object-Oriented Programming (92)		

Personal Summary

- ❖ **First-author NeurIPS 2026 paper (accepted) ICPC Silver Medal** First author of five papers: NeurIPS 2026 (accepted) and four more under review; second author of an ICLR 2027 submission. **Tsinghua University Research Intern** with Prof. Yanyan Lan (06/2026–09/2026). Research spans **RL post-training for LLM agents** (off-policy correction, credit assignment in self-improving multi-agent teams, failure modes of label-free self-play), **constrained RL for molecular design** on measured ADMET panels, and **multi-agent collaboration structures**. Hands-on with **GRPO, LoRA, FSDP-2, vLLM and FP8 on 7B–9B models**; I evaluate with multiple seeds, Holm-corrected tests and pre-registered predictions, and report negative results and boundaries. **Kaggle Featured Bronze Medal (391/6,807)** · 10 pull requests to Tencent open-source agent frameworks.

Research / Project Experience

Accepted at NeurIPS 2026 (Main Track) - **GCD: Correcting Hidden-State Bias in Off-Policy Agentic RL**
First author (co-first, equal contribution, listed first)
2024.11–2026.08

*Partial-rollout RL systems that keep a **stale KV cache** after a policy update do not sample from the old policy: attention reads stale keys and values while the MLP and output layers use new weights. The token-level importance ratio **cannot see this hybrid distribution**: its denominator was stored at rollout time, while its numerator comes from the hybrid pass.*

- ❖ Measured the hidden gap directly on Qwen3.5-9B: median **0.034 nats/token at staleness 256** and **0.108 at 512**. Derived a **per-layer high-probability drift bound** with separate full-attention and Gated DeltaNet terms (tightness 0.64–0.80 on 95.9% of tokens), a **tail-event detector**, drift-triggered **selective recompute**, and a theorem that the corrected weight **composes with the M2PO trust region**.
- ❖ GRPO on 4×H100 (FP8, FSDP-2, verl + vLLM): M2PO+GCD holds **67.1% on MATH-500 at staleness 512 where M2PO collapses to 48.1%**, lifts τ -**bench retail pass**⁴ by **+5.7** at staleness 256 (44.1 vs 38.4, $p < 0.001$), and reaches **1.38× rollout throughput** over full recomputation at staleness 512.
- ❖ In rebuttal, answered the central objection (bias correction or just variance reduction?) with a falsifiable test: the corrected weight is closer to ground-truth recomputation than naive reuse on **68%/74% of tokens** (staleness 256/512) versus 58%/61% for a **matched-mean damping control**; added **Llama-3.1-8B**, stable at 512 where M2PO diverges in 3/3 seeds.

Under Review - **Diversity over Convergence: An Empirical Study of Quality-Diversity Evolution for Multi-Agent Collaboration Structures**
First author
2026.03–2026.07

- ❖ Encoded **agent count, communication topology, per-agent reasoning operators and answer aggregation** as one executable genome and compared **seven search strategies** (MAP-Elites variants, a MaAS-style supernet, AFlow-style MCTS, handcrafted designs and others) under one fitness and budget on Qwen2.5-7B-Instruct (competition MATH, GSM8K; **ten seeds**, Holm-corrected permutation tests).
- ❖ Diverse archives beat AFlow-style MCTS and handcrafted designs on QD-Score (**up to 10×** over handcrafted, $p_{\text{Holm}} = 0.0006$). The central result is a **powered null**: the QD algorithms themselves are indistinguishable (all $p_{\text{Holm}} \geq 0.08$), traced to saturation, the noise floor and **13.2× per-design cost heterogeneity**; in a **7× larger space** they separate ($p_{\text{Holm}} = 0.042$).

Under Review at ICLR 2027 - Learning Deployable Policy Improvements from Measured

Molecular Panels (DeltaADMET-R1)

First author; Tsinghua research internship, supervised by Prof. Yanyan Lan

2026.06–2026.09

*A molecular policy can earn a higher reward yet **retain little of it** once endpoint regression is controlled. Made retained utility, not raw reward, the RL objective for multi-endpoint ADMET optimization over finite candidate panels.*

- ❖ **DeltaADMET-R1**: an exact anchor-relative target with a **non-regression constraint for every endpoint in every context**, and a **total-variation controller** giving a tight outcome-free bound on each endpoint; identity-aligned permutation averaging provably contracts TV; source-conformal and PAC calibration. Trained on Biogen panels; TDC and OpenADMET are used for external evaluation.
- ❖ **CPDPO-R** (contextwise primal-dual, one dual variable per context and endpoint) on Qwen3-8B + LoRA, five seeds, scaffold-disjoint Biogen RLM-solubility panels: retains **about 20% more controlled utility** than matched scalar projection **at the same TV budget** (paired 95% CI excludes zero), and keeps the edge under post hoc 95% source-conformal calibration (193/197 test sources covered). Reported the boundary: on Qwen3-4B and Mistral-7B, higher raw reward did not become a controlled advantage.
- ❖ Located where improvement is lost: a global target sacrifices panels that contextwise constraints protect (**3.14–44.99% of contexts**); identity-aligned averaging lifts certified gain by **about 30%** at unchanged mean gain; a ChemBERTa-Set Transformer baseline slightly beats the LLM target policy after control, which the paper reports.

Under Review at ACL 2027 - Uncertainty Is Not Quality Control: Label-Free Self-Play Drifts

Toward Ill-Posed Problems

First author

2026

*Label-free self-play (R-Zero, TTRL, Absolute Zero) trains a model on problems it is unsure about and treats that agreement band as implicit quality control. It keeps what training has not yet resolved: among model-written problems, mostly the **ill-posed** (unanswerable or ambiguous) ones.*

- ❖ On Qwen3-4B/8B, Phi-4-mini and one Qwen3-14B round, the ill-posed share of admitted problems rises by **11.5–27 points within two rounds**, also under blind re-judging, an open-weight judge and expert human annotation; the **official R-Zero code drifts too** (32.5%→55.5%, 34.0%→61.0%), mostly through its challenger. A class-conditional sharpening law predicts a held-out pool with **MAE 2.4 points**.
- ❖ Qwen3-4B answers unanswerable questions with confidence **29% → 59–76%** of the time while accuracy plateaus. No label-free admission rule we tested stops the drift and rewarding “unknown” backfires, but **test-time abstention works**. **136 pre-registered predictions**, every outcome reported.

Under Review at AAMAS 2027 - Who Taught Whom? Progress-Shapley Credit for Collusion-

Resistant Recursive Self-Improvement of Multi-Agent LLM Teams (ProSPER)

First author

2026

*When proposers, solvers, verifiers and revisers are **all trained with RL** and rewarded for task outcomes, the team can **collude** to inflate reward without learning. Which agent actually made the team learn?*

- ❖ **ProSPER** gives each agent its **Shapley share of the learning progress of the team**: the first-order improvement that the data generated by a coalition induce on a small verifiable anchor set. For a fixed learner state the incentive game is an **exact potential game** with budget-balanced credit; under stated conditions colluding false-accept strategies are **strictly dominated**; a held-out gate bounds regression across rounds.
- ❖ Shared substitution trees and projected LoRA gradients make exact 4-agent Shapley credit cost only **1.31 × rollout**. On Qwen3-8B-Base (four rank-64 LoRA roles, 3 seeds) the average over **13 math, code and general-reasoning benchmarks** rises from **36.3 to 49.9, +4.6** over the strongest self-evolution baseline; with about 1.5k labeled problems it beats RLVR trained on 17k labels by **+3.0**; gains continue for **12 recursive rounds** while every baseline plateaus or regresses by round 5.
- ❖ Under outcome rewards the verifier false-accept rate rises from 31% to 66%; ProSPER keeps it at **5.6%** and raises proposal Vendi diversity by 50%.

Awards & Honors

❖ ICPC International Collegiate Programming Contest	Silver Medal (Team Leader)	2026.04
❖ Kaggle Pokémon TCG AI Battle Challenge (Featured)	Bronze, 391/6,807 (Top 6%)	2026.09
❖ 27th National Robot & AI Competition	National First Prize ×2 (Leader)	2025.08
❖ 14th National Ocean Vehicle Design & Making Competition	National First Prize (Leader)	2025.08
❖ 7th Global Campus AI Algorithm Elite League (2 tracks)	National Second Prize ×2 (Leader)	2025.12
❖ China International College Students Innovation Competition	National Third Prize (Leader)	2025.11
❖ Invention patent · utility model · software copyright	Authorized	2025–2026
❖ In total: 10 national-level and 50 provincial-level awards	Team leader in most	2024–2026